

SAPIENZA
UNIVERSITÀ DI ROMA

Mathematical Analysis 1

Exercises, volume 3 — Limits

The worked exercises on limits, in two parts as the volume of lecture notes, one sheet per chapter, with the solution right after each statement. Limits of sequences: properties of sequences, verifying and computing limits, additional exercises. Limits of functions: limits of real functions of a real variable. The sheets are by Fabio Furini and Gianluca Priori; many exercises are taken from M. Bramanti, *Esercitazioni di Analisi matematica 1*, Esculapio.

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Online version and downloadable material:
fabiofurini.github.io/mathematical-analysis-1/
github.com/fabiofurini/mathematical-analysis-1

Academic year 2026–2027

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Part I

Limits of sequences

Chapter 1

Properties of sequences

Exercise 1.1

Consider the sequence

$$a_n = \log\left(1 + (-1)^n \frac{n}{n+1}\right), \qquad \text{for} \quad n = 1, 2, 3, \ldots$$

Questions:

1. Is the sequence bounded above? If so, determine $\sup\{a_n\}$.
2. Does the sequence have a maximum? If so, determine $\max\{a_n\}$.
3. Is the sequence bounded below? If so, determine $\inf\{a_n\}$.
4. Does the sequence have a minimum? If so, determine $\min\{a_n\}$.

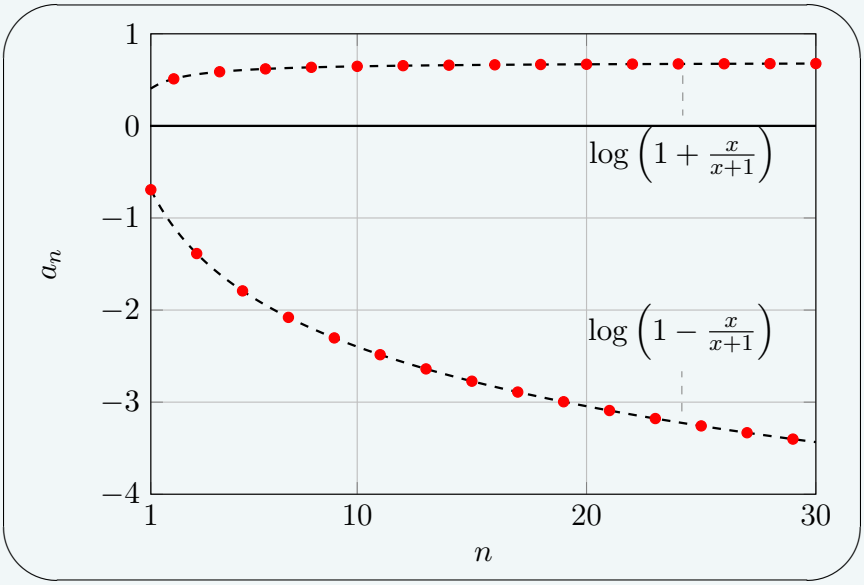
Solution

For $n = 1$ we have

$$\log\left(1 - 1\,\frac{1}{2}\right) = \log\frac{1}{2} \approx -0.69314.$$

For $n = 2$ we have

$$\log\left(1 + 1\,\frac{2}{3}\right) = \log\frac{5}{3} \approx 0.510826.$$



The presence of the alternating sign $(-1)^n$ suggests studying the behavior of the sequence by distinguishing what happens for even n and odd n.

Solution

- If n is even:

$$a_n = \log\left(1+\frac{n}{n+1}\right), \qquad \text{for} \quad n = 2, 4, 6\ldots$$

hence the argument of the logarithm is

$$1+\frac{n}{n+1} = 1+\frac{n+1-1}{n+1} = 2-\frac{1}{n+1}$$

which varies in

$$\left[\frac{5}{3}, 2\right) \quad \text{and the logarithm in} \quad \left[\log\frac{5}{3}, \log 2\right),$$

i.e., it is positive and bounded above. Hence

$$\sup\left\{a_n : n \ \text{ is even } \right\} = \log 2 \approx 0.69314$$

but $\log 2$ is not the maximum of these values.

- If n is odd:

$$a_n = \log\left(1-\frac{n}{n+1}\right), \qquad \text{for} \quad n = 1, 3, 5\ldots$$

hence the argument of the logarithm is

$$1-\frac{n}{n+1} = \frac{1}{n+1}$$

which varies in

$$\left(0, \frac{1}{2}\right] \quad \text{and the logarithm in} \quad \left(-\infty, \log\frac{1}{2}\right],$$

i.e., it is negative and, as n increases, it is unbounded below.

The sequence as a whole is therefore bounded above but not bounded below; it has no maximum (although its sup is finite) and no minimum (because it is unbounded below).

Exercise 1.2

Consider the sequence

$$a_n = e^{-\frac{1}{n}} \sin n, \qquad \text{for } n = 1, 2, 3, \dots$$

Questions:

1. Is the sequence bounded above?
2. Is the sequence bounded below?
3. Is it eventually positive?
4. Does it never vanish?
5. Does it have a limit (finite or infinite)?

Solution

For $n = 1$ we have

$$e^{-1} \sin 1 \approx 0.3679 \cdot 0.8415 \approx 0.309560.$$

For $n = 2$ we have

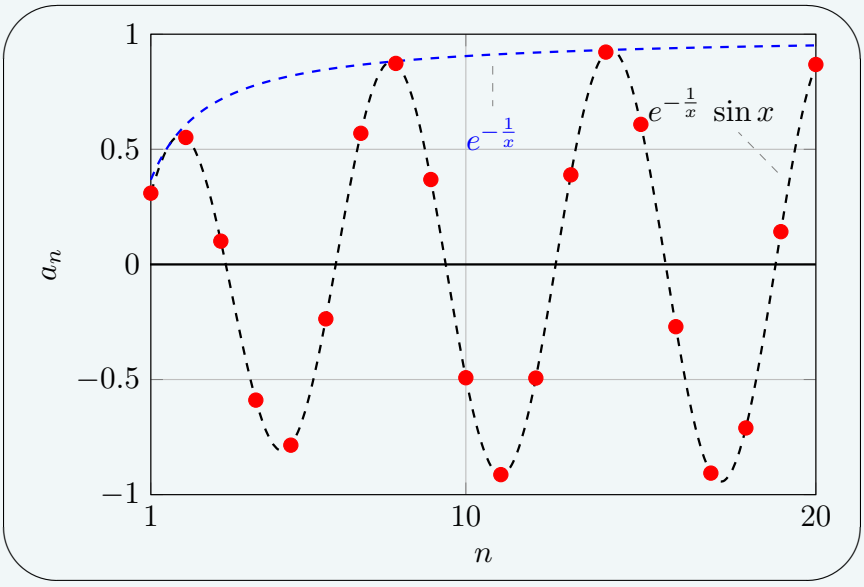
$$e^{-\frac{1}{2}} \sin 2 \approx 0.6065 \cdot 0.9093 \approx 0.551517.$$

For $n = 10$ we have

$$e^{-\frac{1}{10}} \sin 10 \approx 0.9048 \cdot -0.5440 \approx -0.492251.$$

For $n = 20$ we have

$$e^{-\frac{1}{20}} \sin 20 \approx 0.9512 \cdot 0.9129 \approx 0.8684.$$



Solution

The sequence is the product of the sequence

$$n \mapsto e^{-\frac{1}{n}} \quad \text{with} \quad \lim_{n\to+\infty} e^{-\frac{1}{n}} = 1,$$

which is bounded, always nonzero, convergent, and of the sequence

$$n \mapsto \sin n,$$

which is bounded but irregular, and never zero (since n starts from 1, by the irrationality of π, the angle n is never an integer multiple of π).

Hence, the sequence given by the product of these two is bounded, never zero, and irregular.

Since

$$e^{-\frac{1}{n}} \to 1$$

and the sign of $\sin n$ is not eventually constant, the sequence given by the product of the two is not eventually positive.

Exercise 1.3

Consider the sequence

$$a_n = e^n \sin n, \qquad \text{for} \quad n = 1, 2, 3, \dots$$

Questions:

1. Is the sequence bounded above?
2. Is the sequence bounded below?
3. Is it eventually positive?
4. Does it never vanish?
5. Does it have a limit (finite or infinite)?

Solution

For $n = 1$ we have

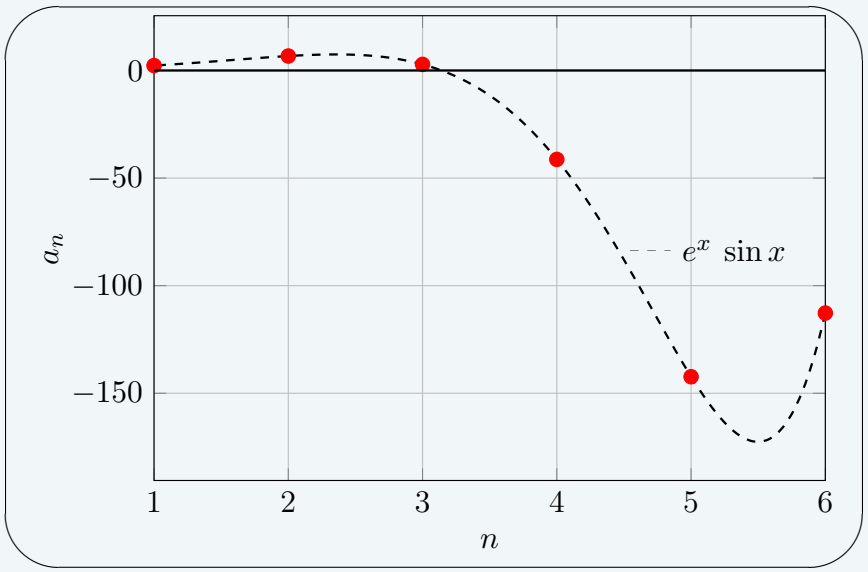
$$e^1 \sin 1 \approx 2.7183 \cdot 0.8415 \approx 2.2874.$$

For $n = 2$ we have

$$e^2 \sin 2 \approx 7.3891 \cdot 0.9093 \approx 6.7188.$$

For $n = 5$ we have

$$e^5 \sin 5 \approx 148.4132 \cdot -0.9589 \approx -142.3170.$$



Solution

The sequence is the product of the sequence

$$n \mapsto e^n \quad \text{with} \quad \lim_{n\to+\infty} e^n = +\infty,$$

which is unbounded above, bounded below, always nonzero, divergent, and of the sequence

$$n \mapsto \sin n,$$

which is bounded but irregular, and never zero (since n starts from 1, by the irrationality of π, the angle n is never an integer multiple of π).

Hence, the sequence given by the product of these two is unbounded above, unbounded below, never zero, and irregular.

Since

$$e^n \to +\infty$$

and the sign of $\sin n$ is not eventually constant, the sequence given by the product of the two is not eventually positive.

Exercise 1.4

Consider the sequence

$$a_n = \frac{n^{(-1)^n}}{n+1}, \qquad \text{for} \quad n = 1, 2, 3, \dots$$

Questions:

1. Is the sequence bounded above? If so, determine $\sup\{a_n\}$.
2. Does the sequence have a maximum? If so, determine $\max\{a_n\}$.

3. Is the sequence bounded below? If so, determine $\inf\{a_n\}$.
4. Does the sequence have a minimum? If so, determine $\min\{a_n\}$.

Solution

For $n = 1$ we have

$$1^{(-1)^1}\frac{1}{2} = 1\,\frac{1}{2}$$

For $n = 2$ we have

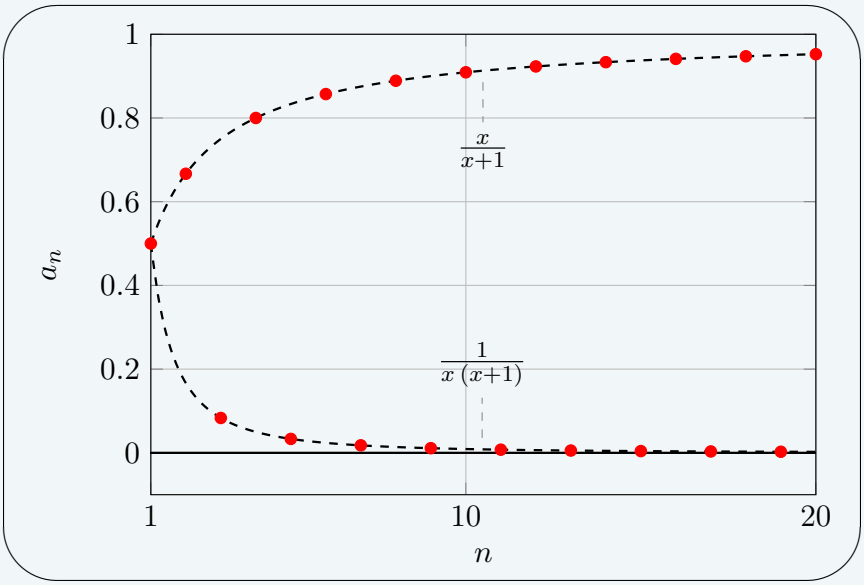
$$2^{(-1)^2}\frac{1}{3} = 2\,\frac{1}{3}.$$

For $n = 3$ we have

$$3^{(-1)^3}\frac{1}{4} = \frac{1}{3}\,\frac{1}{4}.$$

For $n = 4$ we have

$$4^{(-1)^4}\frac{1}{5} = 4\,\frac{1}{5}.$$



The presence of the alternating sign $(-1)^n$ suggests studying the behavior of the sequence by distinguishing what happens for even n and odd n.

Solution

- If n is even:

$$a_n = \frac{n}{n+1}, \qquad \text{for} \quad n = 2, 4, 6\ldots$$

Adding $+1$ and -1 to the numerator, we obtain

$$a_n = \frac{n}{n+1} = \frac{n+1-1}{n+1} = 1 - \frac{1}{n+1}$$

which varies in

$$\left[\frac{2}{3}, 1\right),$$

i.e., it is positive and bounded above. Hence

$$\sup\left\{a_n : n \text{ is even }\right\} = 1$$

but 1 is not the maximum of these values.

- If n is odd:

$$a_n = \frac{\frac{1}{n}}{n+1} = \frac{1}{n} \cdot \frac{1}{n+1} = \frac{1}{n(n+1)}, \qquad \text{for } n = 1, 3, 5 \ldots$$

which varies in

$$\left(0, \frac{1}{2}\right],$$

i.e., it is positive and bounded below. Hence

$$\inf\left\{a_n : n \text{ is odd }\right\} = 0$$

but 0 is not the minimum of these values.

The sequence as a whole is bounded; it has no maximum (although its sup is finite) and no minimum (although its inf is finite).

Chapter 2

Verifying limits of sequences

Exercise 2.1

Using the definition of limit, verify that:

$$\lim_{n\to+\infty} \frac{n}{2n+5} = \frac{1}{2}$$

Solution

We have

$$a_n = \frac{n}{2n+5}$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n > n(\varepsilon) \Rightarrow \frac{1}{2} - \varepsilon < a_n < \frac{1}{2} + \varepsilon$$

We have

$$a_n - \frac{1}{2} = -\frac{5}{4n+10},$$

hence we must verify that

$$n > n(\varepsilon) \Rightarrow -\varepsilon < -\frac{5}{4n+10} < \varepsilon$$

The second inequality is always true, while the first is equivalent to

$$4n + 10 > \frac{5}{\varepsilon} \quad \text{hence it is satisfied for} \quad n > \frac{5}{4\varepsilon} - \frac{5}{2}$$

For a fixed $\varepsilon > 0$, it suffices to choose the first integer

$$n(\varepsilon) > \frac{5}{4\varepsilon} - \frac{5}{2}$$

to satisfy the condition required by the definition of limit.

Exercise 2.2

Using the definition of limit, verify that:

$$\lim_{n\to+\infty} \frac{1}{\sqrt{n+1}} = 0$$

Solution

We have

$$a_n = \frac{1}{\sqrt{n+1}}$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n > n(\varepsilon) \Rightarrow -\varepsilon < a_n < \varepsilon$$

The first inequality is always true, while the second is equivalent to

$$\sqrt{n+1} > \frac{1}{\varepsilon} \quad \text{hence it is satisfied for} \quad n > \frac{1}{\varepsilon^2} - 1$$

For a fixed $\varepsilon > 0$, it suffices to choose the first integer

$$n(\varepsilon) > \frac{1}{\varepsilon^2} - 1$$

to satisfy the condition required by the definition of limit.

Exercise 2.3

Using the definition of limit, verify that:

$$\lim_{n\to+\infty} \log\left(1 + \frac{1}{n}\right) = 0$$

Solution

We have

$$a_n = \log\left(1 + \frac{1}{n}\right)$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n > n(\varepsilon) \Rightarrow -\varepsilon < a_n < \varepsilon$$

The first inequality is always true, while the second is equivalent to

$$1 + \frac{1}{n} < e^{\varepsilon} \quad \text{hence it is satisfied for} \quad n > \frac{1}{e^{\varepsilon} - 1}$$

For a fixed $\varepsilon > 0$, it suffices to choose the first integer

$$n(\varepsilon) > \frac{1}{e^{\varepsilon} - 1}$$

to satisfy the condition required by the definition of limit.

Exercise 2.4

Using the definition of limit, verify that:

$$\lim_{n\to+\infty} \sqrt{4+\frac{1}{n}} = 2$$

Solution

We have

$$a_n = \sqrt{4+\frac{1}{n}}$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n > n(\varepsilon) \Rightarrow 2-\varepsilon < a_n < 2+\varepsilon$$

The first inequality is always true since $a_n > 2$ for every n, while the second is equivalent to

$$4+\frac{1}{n} < 4+\varepsilon^2+4\varepsilon \quad \text{hence it is satisfied for} \quad n > \frac{1}{\varepsilon^2+4\varepsilon}$$

For a fixed $\varepsilon > 0$, it suffices to choose the first integer

$$n(\varepsilon) > \frac{1}{\varepsilon^2+4\varepsilon}$$

to satisfy the condition required by the definition of limit.

Exercise 2.5

Using the definition of limit, verify that:

$$\lim_{n\to+\infty} (n^2-1) = +\infty$$

Solution

We have

$$a_n = n^2-1$$

We must verify that for every $M > 0$ there exists $n(M) \in \mathbb{N}$ such that

$$n > n(M) \Rightarrow a_n > M$$

The inequality

$$n^2-1 > M \quad \text{is satisfied for} \quad n > \sqrt{M+1}$$

Hence, for a fixed $M > 0$, it suffices to choose the first integer

$$n(M) > \sqrt{M+1}$$

to satisfy the required divergence condition.

Exercise 2.6

Using the definition of limit, verify that:

$$\lim_{n\to+\infty} \log\left(\sqrt{n}+1\right) = +\infty$$

Solution

We have

$$a_n = \log\left(\sqrt{n}+1\right)$$

We must verify that for every $M > 0$ there exists $n(M) \in \mathbb{N}$ such that

$$n > n(M) \Rightarrow a_n > M$$

The inequality

$$\log\left(\sqrt{n}+1\right) > M \quad \text{is satisfied for} \quad n > (e^M - 1)^2$$

Hence, for a fixed $M > 0$, it suffices to choose the first integer

$$n(M) > (e^M - 1)^2$$

to satisfy the required divergence condition.

Chapter 3

Computing limits of sequences

3.1 Computing limits with basic techniques

If:

$$a_n \to \ell_a \in \mathbb{R} \quad \text{and} \quad b_n \to \ell_b \in \mathbb{R},$$

we have:

$$a_n \pm b_n \to \ell_a \pm \ell_b, \qquad \frac{a_n}{b_n} \to \frac{\ell_a}{\ell_b} \quad (b_n, \ell_b \neq 0, \text{ eventually}),$$

$$a_n\, b_n \to \ell_a\, \ell_b, \qquad a_n^{b_n} \to {\ell_a}^{\ell_b} \quad (a_n, \ell_a > 0, \text{ eventually}).$$

If:

$$a_n \to \ell_a \in \mathbb{R}, \quad b_n \to +\infty \quad \text{and} \quad c_n \to +\infty,$$

we have:

$$a_n + b_n \to \ell_a + \infty = +\infty, \quad a_n - b_n \to \ell_a - \infty = -\infty,$$

$$b_n + c_n \to +\infty + \infty = +\infty, \quad -b_n - c_n \to -\infty - \infty = -\infty.$$

If:

$$a_n \to \ell_a \in \mathbb{R}, \quad b_n \to 0^+ \text{ or } b_n \to 0^- \quad \text{and} \quad c_n \to \infty,$$

we have:

$$a_n\ c_n \to \ell_a\ \infty = \infty \ \ (\ell_a \neq 0), \qquad \frac{a_n}{b_n} \to \frac{\ell_a}{0} = \infty \ \ (\ell_a \neq 0), \qquad \frac{a_n}{c_n} \to \frac{\ell_a}{\infty} = 0.$$

The hypothesis that $\{b_n\}$ tends to zero from above or from below (and hence that it eventually has constant sign, with $b_n \neq 0$) is necessary: with $a_n = 1$ and $b_n = \frac{(-1)^n}{n}$ we have $\frac{a_n}{b_n} = (-1)^n\, n$, which is irregular.

The four main indeterminate forms are:

$$[+\infty - \infty], \quad [0 \cdot \infty], \quad \left[\frac{0}{0}\right] \quad \text{and} \quad \left[\frac{\infty}{\infty}\right].$$

There are also three other indeterminate forms derived from $[0 \cdot \infty]$:

$$[1^{\infty}], \quad [0^0] \quad \text{and} \quad [\infty^0].$$

We have:

$$\lim_{n\to+\infty} n^\alpha = \begin{cases} +\infty & \text{if } \alpha > 0 \\ 1 & \text{if } \alpha = 0 \\ 0 & \text{if } \alpha < 0 \end{cases} \qquad \lim_{n\to+\infty} a^n = \begin{cases} +\infty & \text{if } a > 1 \\ 1 & \text{if } a = 1 \\ 0 & \text{if } |a| < 1 \\ \text{does not exist} & \text{if } a \le -1 \end{cases}$$

Exercise 3.1

Compute the following limit, which is an (elementary) indeterminate form:

$$\lim_{n\to+\infty} \frac{n^2+2n}{n+1} = \left[\frac{\infty}{\infty}\right]$$

Solution

$$\lim_{n\to+\infty} \frac{n^2+2n}{n+1} = \lim_{n\to+\infty} \frac{\overbrace{n^2\left(1+\overbrace{\frac{2}{n}}^{\to 0}\right)}^{\to 1}}{\underbrace{n\left(1+\underbrace{\frac{1}{n}}_{\to 0}\right)}_{\to 1}} = \lim_{n\to+\infty} \frac{n^2}{n} = \lim_{n\to+\infty} n = +\infty.$$

Exercise 3.2

Compute the following limit, which is an (elementary) indeterminate form:

$$\lim_{n\to+\infty} \frac{n^4+5}{n^5+7n-1} = \left[\frac{\infty}{\infty}\right]$$

Solution

$$\lim_{n\to+\infty} \frac{n^4+5}{n^5+7n-1} = \lim_{n\to+\infty} \frac{n^4\left(1+\frac{5}{n^4}\right)}{n^5\left(1+\frac{7}{n^4}-\frac{1}{n^5}\right)} = \lim_{n\to+\infty} \frac{n^4}{n^5} = \lim_{n\to+\infty} \frac{1}{n} = 0.$$

Exercise 3.3

Compute the following limit, which is an (elementary) indeterminate form:

$$\lim_{n\to+\infty} \frac{1-n^2}{(n+2)^2} = \left[\frac{\infty}{\infty}\right]$$

Solution

$$\lim_{n\to+\infty} \frac{1-n^2}{(n+2)^2} = \lim_{n\to+\infty} \frac{-n^2\left(1-\frac{1}{n^2}\right)}{n^2\left(1+\frac{4}{n}+\frac{4}{n^2}\right)} = \lim_{n\to+\infty} \frac{-n^2}{n^2} = \lim_{n\to+\infty} -1 = -1.$$

Exercise 3.4

Compute the following limit, which is an (elementary) indeterminate form:

$$\lim_{n\to+\infty} \sqrt{n^2+1}-\sqrt{n} = [+\infty-\infty]$$

Solution

$$\lim_{n\to+\infty} \sqrt{n^2+1}-\sqrt{n} = \lim_{n\to+\infty} n\left(\underbrace{\sqrt{1+\underbrace{\frac{1}{n^2}}_{\to 0}}}_{\to 1} - \underbrace{\sqrt{\frac{1}{n}}}_{\to 0}\right) = \lim_{n\to+\infty} n = +\infty.$$

Solution

Alternative method: multiplying and dividing by $\sqrt{n^2+1}+\sqrt{n}$ we obtain

$$\frac{(\sqrt{n^2+1}-\sqrt{n})(\sqrt{n^2+1}+\sqrt{n})}{\sqrt{n^2+1}+\sqrt{n}} = \frac{\left(\sqrt{n^2+1}\right)^2-(\sqrt{n})^2}{\sqrt{n^2+1}+\sqrt{n}} = \frac{n^2+1-n}{\sqrt{n^2+1}+\sqrt{n}},$$

recalling that $(a-b)(a+b)=a^2-b^2$. Moreover, we have

$$\frac{n^2+1-n}{\sqrt{n^2+1}+\sqrt{n}} = \frac{n^2\left(1+\frac{1}{n^2}-\frac{1}{n}\right)}{\sqrt{n^2\left(1+\frac{1}{n^2}\right)}+\sqrt{n}} = \frac{n^2\left(1+\frac{1}{n^2}-\frac{1}{n}\right)}{n\sqrt{\left(1+\frac{1}{n^2}\right)}+n^{\frac{1}{2}}} = \frac{n^2\left(1+\frac{1}{n^2}-\frac{1}{n}\right)}{n\left(\sqrt{\left(1+\frac{1}{n^2}\right)}+\frac{1}{n^{1/2}}\right)}.$$

Hence

$$\lim_{n\to+\infty}\sqrt{n^2+1}-\sqrt{n} = \lim_{n\to+\infty} n\,\frac{\overbrace{\left(1+\frac{1}{n^2}-\frac{1}{n}\right)}^{\to 1}}{\underbrace{\left(\sqrt{\left(1+\frac{1}{n^2}\right)}+\frac{1}{n^{1/2}}\right)}_{\to 1}} = \lim_{n\to+\infty} n = +\infty.$$

Exercise 3.5

Compute the following limit, which is an (elementary) indeterminate form:

$$\lim_{n\to+\infty} e^n - 2^n = [+\infty-\infty]$$

Solution

$$\lim_{n\to+\infty} e^n-2^n = \lim_{n\to+\infty} e^n \underbrace{\left(1-\underbrace{\left(\frac{2}{e}\right)^n}_{\to 0}\right)}_{\to 1} = \lim_{n\to+\infty} e^n = +\infty$$

since

$$e^n \to +\infty \quad \text{and} \quad \left(\frac{2}{e}\right)^n \to 0.$$

Exercise 3.6

Compute the following limit, which is an (elementary) indeterminate form:

$$\lim_{n\to+\infty} 3^n+4^n-5^n = [+\infty+\infty-\infty] = [+\infty-\infty]$$

Solution

$$\lim_{n\to+\infty} 3^n + 4^n - 5^n = \lim_{n\to+\infty} -5^n \underbrace{\left(1 - \underbrace{\left(\frac{3}{5}\right)^n}_{\to 0} - \underbrace{\left(\frac{4}{5}\right)^n}_{\to 0}\right)}_{\to 1} = \lim_{n\to+\infty} -5^n = -\infty$$

since

$$-5^n \to -\infty, \quad \left(\frac{3}{5}\right)^n \to 0 \quad \text{and} \quad \left(\frac{4}{5}\right)^n \to 0.$$

Exercise 3.7

Compute the following simple limit

$$\lim_{n\to+\infty} n^{\sqrt{2}}$$

Solution

For every $\alpha > 0$, even irrational, $\lim_{n\to+\infty} n^\alpha = +\infty$.

Exercise 3.8

Compute the following simple limit

$$\lim_{n\to+\infty} n^{-e}$$

Solution

$$\lim_{n\to+\infty} n^{-e} = \lim_{n\to+\infty} \frac{1}{n^e} = 0.$$

Exercise 3.9

Compute the following simple limit

$$\lim_{n\to+\infty} \left(\frac{3\,n+5}{n^2+120}\right)^{2\,n}$$

Solution

We have

$$\frac{3\,n+5}{n^2+120} = \frac{3\,n\left(1+\frac{5}{3\,n}\right)}{n^2\left(1+\frac{120}{n^2}\right)} = \frac{3}{n}\,\frac{\left(1+\frac{5}{3\,n}\right)}{\left(1+\frac{120}{n^2}\right)} \to 0 \quad \text{and} \quad 2\,n \to +\infty.$$

Hence

$$\lim_{n\to+\infty}\left(\frac{3\,n+5}{n^2+120}\right)^{2\,n}=0^{+\infty}=0.$$

Exercise 3.10

Compute the following simple limit

$$\lim_{n\to+\infty}\left(\frac{10\,n^2-5\,n}{n^3-8}\right)^{-3\,n+8}$$

Solution

We have

$$\frac{10\,n^2-5\,n}{n^3-8}=\frac{10\,n^2\left(1-\frac{5\,n}{10\,n^2}\right)}{n^3\left(1-\frac{8}{n^3}\right)}=\frac{10}{n}\,\frac{\left(1-\frac{1}{2\,n}\right)}{\left(1-\frac{8}{n^3}\right)}\to 0\quad\text{and}\quad -3\,n+8\to-\infty.$$

Hence

$$\lim_{n\to+\infty}\left(\frac{10\,n^2-5\,n}{n^3-8}\right)^{-3\,n+8}=0^{-\infty}=\frac{1}{0^{+\infty}}=\frac{1}{0}=+\infty.$$

3.2 Computing limits using the hierarchy of infinities

For every $a>1$ and $\alpha>0$ we have:

$$\lim_{n\to+\infty}\frac{\log_a n}{n^\alpha}=0,\qquad \lim_{n\to+\infty}\frac{n^\alpha}{a^n}=0,$$

$$\lim_{n\to+\infty}\frac{a^n}{n!}=0,\qquad \lim_{n\to+\infty}\frac{n!}{n^n}=0.$$

Hence the following infinities are listed in increasing order:

$$\log n,\quad n^\alpha,\quad a^n,\quad n!,\quad n^n.$$

Exercise 3.11

Compute the limit of the following sequence, if it exists

$$a_n=\frac{2^{1/n}+n^2+3^{-n}}{\log^6 n+2+n}$$

Solution

Consider the numerator; we have:

$$2^{1/n} \to 1, \quad n^2 \to +\infty, \quad 3^{-n} \to 0$$

The power of n with the highest degree is n^2, and moreover we have

$$\frac{3^{-n}}{n^2} = \frac{1}{3^n\, n^2} \quad \text{hence} \quad \frac{3^{-n}}{n^2} \to 0 \quad \text{and} \quad \frac{2^{1/n}}{n^2} \to 0.$$

Consider the denominator; we have:

$$\log^6 n \to +\infty, \quad 2 \to 2, \quad n \to +\infty$$

By the hierarchy of infinities theorem, we have

$$\frac{\log^6 n}{n} = \left(\underbrace{\frac{\log n}{n^{1/6}}}_{\to 0} \right)^6 \to 0.$$

We therefore factor out n^2 in the numerator and n in the denominator and obtain:

$$\frac{n^2 \left(1 + \frac{2^{1/n}}{n^2} + \frac{3^{-n}}{n^2}\right)}{n \left(1 + \frac{\log^6 n}{n} + \frac{2}{n}\right)} = n \, \frac{\left(1 + \overbrace{\frac{2^{1/n}}{n^2}}^{\to 0} + \overbrace{\frac{3^{-n}}{n^2}}^{\to 0}\right)}{\left(1 + \underbrace{\frac{\log^6 n}{n}}_{\to 0} + \underbrace{\frac{2}{n}}_{\to 0}\right)}$$

$$\lim_{n\to+\infty} \frac{2^{1/n} + n^2 + 3^{-n}}{\log^6 n + 2 + n} = \lim_{n\to+\infty} n \, \frac{\overbrace{\left(1 + \frac{2^{1/n}}{n^2} + \frac{3^{-n}}{n^2}\right)}^{\to 1}}{\left(1 + \frac{\log^6 n}{n} + \frac{2}{n}\right)} = +\infty$$

Exercise 3.12

Compute the limit of the following sequence, if it exists

$$a_n = \frac{3\, n^3 + 3^n + \log n}{\log^6 n + 2^{2\, n} + n^5}$$

Solution

Consider the numerator; we have:

$$3\, n^3 \to +\infty, \quad 3^n \to +\infty, \quad \log n \to +\infty$$

that is, a sum of sequences tending to infinity. We also have:

$$\frac{3\,n^3}{3^n} \to 0 \quad \text{and} \quad \frac{\log n}{3^n} \to 0,$$

by the hierarchy of infinities theorem. Hence the principal part is 3^n. Consider the denominator; we have:

$$\log^6 n \to +\infty, \quad 2^{2\,n} = 4^n \to +\infty, \quad n^5 \to +\infty$$

that is, a sum of sequences tending to infinity. We also have:

$$\frac{\log^6 n}{4^n} \to 0 \quad \text{and} \quad \frac{n^5}{4^n} \to 0,$$

by the hierarchy of infinities theorem. Hence the principal part is 4^n.

We therefore factor out 3^n in the numerator and 4^n in the denominator and obtain:

$$\frac{3^n\left(1+\frac{3\,n^3}{3^n}+\frac{\log n}{3^n}\right)}{4^n\left(1+\frac{\log^6 n}{4^n}+\frac{n^5}{4^n}\right)} = \left(\frac{3}{4}\right)^n \frac{\left(1+\overbrace{\frac{3\,n^3}{3^n}}^{\to 0}+\overbrace{\frac{\log n}{3^n}}^{\to 0}\right)}{\left(1+\underbrace{\frac{\log^6 n}{4^n}}_{\to 0}+\underbrace{\frac{n^5}{4^n}}_{\to 0}\right)}$$

hence

$$\lim_{n\to+\infty}\frac{3\,n^3+3^n+\log n}{\log^6 n+2^{2\,n}+n^5} = \lim_{n\to+\infty}\left(\frac{3}{4}\right)^n \underbrace{\frac{\left(1+\frac{3\,n^3}{3^n}+\frac{\log n}{3^n}\right)}{\left(1+\frac{\log^6 n}{4^n}+\frac{n^5}{4^n}\right)}}_{\to 1} = 0$$

Exercise 3.13

Compute the following simple limit

$$\lim_{n\to+\infty}\sqrt[n]{n^2}$$

Solution

We have

$$\sqrt[n]{n} = n^{\frac{1}{n}} = e^{\log n^{1/n}} = e^{\frac{\log n}{n}}$$

and, studying the sequence in the exponent,

$$a_n = \frac{\log n}{n} \quad \text{we have} \quad a_n \to 0$$

thanks to the hierarchy of infinities theorem. Hence we have:

$$\lim_{n\to+\infty} \sqrt[n]{n} = \lim_{n\to+\infty} e^{\frac{\log n}{n}} = 1.$$

Consequently

$$\lim_{n\to+\infty} \sqrt[n]{n^2} = \lim_{n\to+\infty} \left(\sqrt[n]{n}\right)^2 = 1^2 = 1.$$

3.3 Computing limits using asymptotic estimates

We have $a_n \sim b_n$ if $\frac{a_n}{b_n} \to 1$ or if $a_n = b_n\, c_n$ with $c_n \to 1$.

Moreover, we have $a_n + b_n \sim a_n$ if $\frac{b_n}{a_n} \to 0$

Exercise 3.14

Compute the following limit

$$\lim_{n\to+\infty} \frac{2^n + n^2 + 1}{5^n + 2^n + n}$$

Solution

We have

$$2^n + n^2 + 1 \sim 2^n \quad \text{since} \quad \frac{n^2}{2^n} \to 0 \quad \textit{and} \quad \frac{1}{2^n} \to 0$$

and

$$5^n + 2^n + n \sim 5^n \quad \text{since} \quad \frac{2^n}{5^n} = \left(\frac{2}{5}\right)^n \to 0 \quad \textit{and} \quad \frac{n}{5^n} \to 0$$

hence

$$\lim_{n\to+\infty} \frac{2^n + n^2 + 1}{5^n + 2^n + n} = \lim_{n\to+\infty} \frac{2^n}{5^n} = \lim_{n\to+\infty} \left(\frac{2}{5}\right)^n = 0$$

Exercise 3.15

Compute the following limit

$$\lim_{n\to+\infty} \frac{n! - 5^n}{7^n}$$

Solution

We have

$$n! - 5^n \sim n! \quad \text{since} \quad \frac{5^n}{n!} \to 0$$

hence

$$\lim_{n\to+\infty} \frac{n! - 5^n}{7^n} = \lim_{n\to+\infty} \frac{n!}{7^n} = +\infty$$

Exercise 3.16

Compute the following limit

$$\lim_{n\to+\infty} \frac{2^{3n-1} - n^2}{(2n)! - 5^n}$$

Solution

We have

$$2^{3n-1} - n^2 \sim 2^{3n-1} \quad \text{since} \quad -\frac{n^2}{2^{3n-1}} \to 0$$

and

$$(2n)! - 5^n \sim (2n)! \quad \text{since} \quad -\frac{5^n}{(2n)!} \to 0$$

hence

$$\lim_{n\to+\infty} \frac{2^{3n-1} - n^2}{(2n)! - 5^n} = \lim_{n\to+\infty} \frac{2^{3n-1}}{(2n)!} = 0$$

Exercise 3.17

Compute the limit of the following sequence, if it exists

$$\lim_{n\to+\infty} \frac{3^n}{n!\, n^{1/n}}$$

Solution

We have

$$n!\, n^{1/n} \sim n! \quad \text{since} \quad n^{1/n} = \sqrt[n]{n} \to 1$$

because

$$n^{1/n} = e^{\log n^{1/n}} = e^{\frac{\log n}{n}} \quad \text{and} \quad \frac{\log n}{n} \to 0.$$

Hence

$$\lim_{n\to+\infty} \frac{3^n}{n!\, n^{1/n}} = \lim_{n\to+\infty} \frac{3^n}{n!} = 0.$$

Exercise 3.18

Compute the limit of the following sequence, if it exists

$$\lim_{n\to+\infty} \frac{\log(1 + e^n)}{\sqrt{1 + n^2}}$$

Solution

Observing that

$$\log(1+e^n) \sim \log(e^n) = n\log e = n$$

we have

$$\lim_{n\to+\infty} \frac{\log(1+e^n)}{\sqrt{1+n^2}} = \lim_{n\to+\infty} \frac{n}{\sqrt{1+n^2}} = \lim_{n\to+\infty} \sqrt{\frac{n^2}{1+n^2}} = 1$$

because

$$\frac{n^2}{1+n^2} \to 1.$$

Exercise 3.19

Compute the limit of the following sequence, if it exists

$$\lim_{n\to+\infty} \frac{n+1}{\log_2(3+n^n)}$$

Solution

Observing that

$$\log_2(3+n^n) \sim \log_2(n^n) = n\log_2 n$$

we have

$$\lim_{n\to+\infty} \frac{n+1}{\log_2(3+n^n)} = \lim_{n\to+\infty} \frac{n+1}{n\log_2 n} = 0$$

because

$$\frac{n+1}{n} \to 1 \quad \text{and} \quad \frac{1}{\log_2 n} \to 0.$$

3.4 Computing limits with the sequence converging to Euler's number e

For indeterminate forms $1^{\pm\infty}$, one can use:

$$\left(1+\frac{1}{a_n}\right)^{a_n} \to e \quad \text{as } a_n \to \pm\infty.$$

Exercise 3.20

Compute the following limit

$$\lim_{n\to+\infty} \left(\frac{n-1}{n-3}\right)^n$$

Solution

We have

$$\left(\frac{n-1}{n-3}\right)^n = \left(\frac{n-3-1+3}{n-3}\right)^n = \left(1+\frac{2}{n-3}\right)^n = \left(1+\frac{1}{\frac{n-3}{2}}\right)^n$$

hence

$$\lim_{n\to+\infty}\left(\frac{n-1}{n-3}\right)^n = \lim_{n\to+\infty}\left[\left(1+\frac{1}{\frac{n-3}{2}}\right)^{\frac{n-3}{2}}\right]^{\frac{2n}{n-3}} = e^2,$$

since

$$\left(1+\frac{1}{\frac{n-3}{2}}\right)^{\frac{n-3}{2}} \to e, \quad 2\left(\frac{n}{n-3}\right) = 2\left(\frac{n-3+3}{n-3}\right) = 2\left(1+\underbrace{\frac{3}{n-3}}_{\to 0}\right) \to 2$$

and

$$\frac{n-3}{2} \to +\infty.$$

Exercise 3.21

Compute the following limit

$$\lim_{n\to+\infty}\left(\frac{n^2+1}{n^2}\right)^n$$

Solution

$$\lim_{n\to+\infty}\left(\frac{n^2+1}{n^2}\right)^n = \lim_{n\to+\infty}\left[\left(1+\frac{1}{n^2}\right)^{n^2}\right]^{\frac{1}{n}} = e^0 = 1$$

since

$$\left(1+\frac{1}{n^2}\right)^{n^2} \to e, \quad \frac{1}{n} \to 0$$

and

$$n^2 \to +\infty.$$

Exercise 3.22

Compute the following limit

$$\lim_{n\to+\infty}\left(\frac{2n-5}{2n}\right)^{-n}$$

Solution

$$\lim_{n\to+\infty}\left(\frac{2n-5}{2n}\right)^{-n}=\lim_{n\to+\infty}\left[\left(1+\frac{1}{-\frac{2n}{5}}\right)^{-\frac{2n}{5}}\right]^{\frac{5}{2}}=\sqrt{e^5}.$$

since

$$\left(1+\frac{1}{-\frac{2n}{5}}\right)^{-\frac{2n}{5}}\to e \quad \text{and} \quad -\frac{2n}{5}\to-\infty.$$

Exercise 3.23

Compute the following limit

$$\lim_{n\to+\infty}\frac{n^{n-1}}{(n-1)^n}$$

Solution

We have

$$\frac{n^{n-1}}{(n-1)^n}=\frac{1}{n}\left(\frac{n}{n-1}\right)^n=\frac{1}{n}\left(\frac{n-1}{n}\right)^{-n}=\frac{1}{n}\left(1+\frac{1}{-n}\right)^{-n}$$

hence

$$\lim_{n\to+\infty}\frac{n^{n-1}}{(n-1)^n}=\lim_{n\to+\infty}\frac{1}{n}\left(1+\frac{1}{-n}\right)^{-n}=0$$

since

$$\left(1+\frac{1}{-n}\right)^{-n}\to e \quad \text{and} \quad \frac{1}{n}\to 0.$$

Exercise 3.24

Compute the limit of the following sequence, if it exists

$$\lim_{n\to+\infty}\left(\frac{n^2+2}{n^2+n+1}\right)^{2\,n}$$

Solution

Since

$$\left(\frac{n^2+2}{n^2+n+1}\right)\to 1 \quad \text{and} \quad 2\,n\to+\infty$$

we have an indeterminate form of type $[1^{+\infty}]$.

Since

$$\left(\frac{n^2+2}{n^2+n+1}\right)^{2\,n} = \left(\frac{n^2+n+1-n+1}{n^2+n+1}\right)^{2\,n} = \left(1+\frac{-n+1}{n^2+n+1}\right)^{2\,n},$$

we can therefore write

$$\left(1+\frac{-n+1}{n^2+n+1}\right)^{2\,n} = \left(1+\frac{1}{\frac{n^2+n+1}{-n+1}}\right)^{2\,n} = \left[\left(1+\frac{1}{\frac{n^2+n+1}{-n+1}}\right)^{\frac{n^2+n+1}{-n+1}}\right]^{\left(\frac{-n+1}{n^2+n+1}\right)2\,n}.$$

Hence

$$\lim_{n\to+\infty}\left(\frac{n^2+2}{n^2+n+1}\right)^{2\,n} = \lim_{n\to+\infty}\left[\left(1+\frac{1}{\frac{n^2+n+1}{-n+1}}\right)^{\frac{n^2+n+1}{-n+1}}\right]^{\left(\frac{-n+1}{n^2+n+1}\right)2\,n} = e^{-2},$$

since

$$\left(1+\frac{1}{\frac{n^2+n+1}{-n+1}}\right)^{\frac{n^2+n+1}{-n+1}} \to e, \quad \left(\frac{-n+1}{n^2+n+1}\right)2\,n = \left(\frac{\overbrace{-2\,n^2+2\,n}^{\sim -2\,n^2}}{\underbrace{n^2+n+1}_{\sim n^2}}\right) \to -2$$

and

$$\frac{n^2+n+1}{-n+1} = \frac{n^2\overbrace{\left(1+\frac{1}{n}+\frac{1}{n^2}\right)}^{\to 1}}{-n\underbrace{\left(1-\frac{1}{n}\right)}_{\to 1}} \to -\infty.$$

3.5 Computing limits with the comparison theorem

- If $c_n \to 0$ and $|b_n| \le c_n$ eventually, then $b_n \to 0$.
- If $c_n \to 0$ and b_n is bounded (even if not convergent), then $c_n\,b_n \to 0$. The product of an infinitesimal sequence and a bounded one is infinitesimal.
- A sum of infinities and bounded sequences is asymptotically equivalent to the infinity of highest order.

Exercise 3.25

Compute the following limit

$$\lim_{n\to+\infty}\frac{n+\sin n}{\cos n+\log n}$$

Solution

We have

$$n + \sin n \sim n \quad \text{since} \quad \frac{\sin n}{n} \to 0$$

the product of an infinitesimal sequence $\{\frac{1}{n}\}$ and a bounded one $\{\sin n\}$. Moreover,

$$\cos n + \log n \sim \log n \quad \text{since} \quad \frac{\cos n}{\log n} \to 0$$

the product of an infinitesimal sequence $\left\{\frac{1}{\log n}\right\}$ and a bounded one $\{\cos n\}$. Hence

$$\lim_{n\to+\infty} \frac{n + \sin n}{\cos n + \log n} = \lim_{n\to+\infty} \frac{n}{\log n} = +\infty$$

Exercise 3.26

Compute the limit of the following sequence, if it exists

$$\lim_{n\to+\infty} \frac{n\ \sin n + \sin n^2}{n^2 + 1}$$

Solution

We have

$$\frac{n\ \sin n + \sin n^2}{n^2 + 1} = \frac{n\ \sin n}{n^2 + 1} + \frac{\sin n^2}{n^2 + 1}$$

and

$$\frac{n}{n^2 + 1}\ \sin n \to 0 \quad \text{since} \quad \frac{n}{n^2 + 1} \to 0 \quad \textit{and} \quad \sin n \quad \text{is bounded.}$$

Moreover

$$\frac{1}{n^2 + 1}\ \sin n^2 \to 0 \quad \text{since} \quad \frac{1}{n^2 + 1} \to 0 \quad \textit{and} \quad \sin n^2 \quad \text{is bounded.}$$

Hence

$$\lim_{n\to+\infty} \frac{n\ \sin n + \sin n^2}{n^2 + 1} = \lim_{n\to+\infty} \underbrace{\frac{n\ \sin n}{n^2 + 1}}_{\to 0} + \underbrace{\frac{\sin n^2}{n^2 + 1}}_{\to 0} = 0$$

Solution

Alternative method.

Note that the sequences $\{\sin n\}$ and $\{\sin n^2\}$ are irregular but bounded. Hence we can write the following upper bound

$$\left|\frac{n\ \sin n + \sin n^2}{n^2 + 1}\right| \leq \frac{n + 1}{n^2 + 1}$$

and we have

$$\lim_{n\to+\infty} \frac{n+1}{n^2+1} = \lim_{n\to+\infty} \frac{n}{n^2} = \lim_{n\to+\infty} \frac{1}{n} = 0.$$

Hence, by the comparison theorem, we have:

$$\lim_{n\to+\infty} \frac{n \sin n + \sin n^2}{n^2+1} = 0$$

3.6 Computing limits with the ratio test

Given a positive sequence ($a_n > 0$ for every n):

$$\text{if there exists } \lim_{n\to+\infty} \frac{a_{n+1}}{a_n} = \ell \text{ and } \ell < 1, \text{ then } a_n \to 0,$$

$$\text{if there exists } \lim_{n\to+\infty} \frac{a_{n+1}}{a_n} = \ell \text{ and } \ell > 1 \text{ (or } \ell = +\infty), \text{ then } a_n \to +\infty.$$

Exercise 3.27

Compute the limit of the following sequence, if it exists

$$\lim_{n\to+\infty} \frac{3^{n-1}}{n\,(n+1)!}$$

Solution

Since this is a sequence with positive terms whose general term is the ratio of simpler sequences of exponential and factorial type, it is natural to apply the ratio test.

We therefore compute:

$$\frac{a_{n+1}}{a_n} = \frac{\frac{3^n}{(n+1)\cdot(n+2)!}}{\frac{3^{n-1}}{n\,(n+1)!}} = \frac{3^n}{(n+1)\cdot \underbrace{(n+2)!}_{=(n+1)!\cdot(n+2)}} \frac{n\,(n+1)!}{3^{n-1}} =$$

$$= \frac{3\,n}{(n+1)(n+2)} = \frac{3\,n}{n^2+2\,n+n+2} = \frac{3\,n}{n^2\left(1+\frac{3}{n}+\frac{2}{n^2}\right)} = \frac{3}{n\left(1+\frac{3}{n}+\frac{2}{n^2}\right)}$$

We have:

$$\lim_{n\to+\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to+\infty} \frac{3}{n\left(1+\frac{3}{n}+\frac{2}{n^2}\right)} = 0$$

Hence, by the ratio test:

$$\lim_{n\to+\infty} \frac{3^{n-1}}{n\,(n+1)!} = 0$$

Exercise 3.28

Compute the limit of the following sequence, if it exists

$$\lim_{n\to+\infty} \frac{n^3\, 2^n}{n!}$$

Solution

We have a positive sequence; we compute:

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^3\, 2^{(n+1)}}{(n+1)!}\frac{n!}{n^3\, 2^n} = \underbrace{\left(\frac{n+1}{n}\right)^3}_{\to 1}\frac{2}{n+1} \sim \frac{2}{n+1}$$

We therefore have:

$$\lim_{n\to+\infty}\frac{a_{n+1}}{a_n} = \lim_{n\to+\infty}\frac{2}{n+1} = 0$$

Then, by the ratio test:

$$\lim_{n\to+\infty} \frac{n^3\, 2^n}{n!} = 0$$

Exercise 3.29

Compute the limit of the following sequence, if it exists

$$\lim_{n\to+\infty} \frac{n^n}{(n+1)!}$$

Solution

We have a positive sequence; we compute:

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^{n+1}}{(n+2)!}\frac{(n+1)!}{n^n} = \underbrace{\frac{n+1}{n+2}}_{\to 1}\left(\frac{n+1}{n}\right)^n \sim \left(1+\frac{1}{n}\right)^n$$

We therefore have:

$$\lim_{n\to+\infty}\frac{a_{n+1}}{a_n} = \lim_{n\to+\infty}\left(1+\frac{1}{n}\right)^n = e > 1$$

Then, by the ratio test:

$$\lim_{n\to+\infty} \frac{n^n}{(n+1)!} = +\infty$$

Exercise 3.30

Compute the limit of the following sequence, if it exists

$$\lim_{n\to+\infty} \frac{n^{\frac{1}{n}}\, 2^n}{(n+1)!}$$

Solution

We have:

$$\frac{\overbrace{n^{\frac{1}{n}}}^{\to 1}\, 2^n}{(n+1)!} \sim \frac{2^n}{(n+1)!} \equiv b_n$$

We now have the positive sequence $\{b_n\}$; we compute:

$$\frac{b_{n+1}}{b_n} = \frac{2^{n+1}}{(n+2)!}\frac{(n+1)!}{2^n} = \frac{2}{n+2}$$

We therefore have:

$$\lim_{n\to+\infty} \frac{b_{n+1}}{b_n} = \lim_{n\to+\infty} \frac{2}{n+2} = 0$$

Then, by the ratio test:

$$\lim_{n\to+\infty} \frac{n^{\frac{1}{n}}\, 2^n}{(n+1)!} = 0$$

3.7 Discussion depending on a parameter

Exercise 3.31

Compute, as the parameter $\alpha \in \mathbb{R}$ varies, the limit of the sequence

$$\lim_{n\to+\infty} \frac{5^{3\alpha n}}{2^{3n+1}}$$

Solution

We have

$$\frac{5^{3\alpha n}}{2^{3n+1}} = \frac{1}{2}\left(\frac{5^\alpha}{2}\right)^{3n}$$

hence

$$\lim_{n\to+\infty} \frac{5^{3\alpha n}}{2^{3n+1}} = \lim_{n\to+\infty} \frac{1}{2}\left(\frac{5^\alpha}{2}\right)^{3n} = \frac{1}{2}\lim_{n\to+\infty}\left(\frac{5^\alpha}{2}\right)^{3n} = \ell$$

At this point, we can study three cases:

- *Case 1*

$$\frac{5^\alpha}{2} > 1 \iff \alpha > \log_5 2$$

then $\ell = +\infty$;

- *Case 2*

$$0 < \frac{5^\alpha}{2} < 1 \iff \alpha < \log_5 2$$

then $\ell = 0$;

- *Case 3*

$$\frac{5^\alpha}{2} = 1 \iff \alpha = \log_5 2$$

then $\ell = \frac{1}{2}$.

In conclusion:

$$\lim_{n\to+\infty} \frac{5^{3\alpha n}}{2^{3n+1}} = \begin{cases} +\infty & \text{if } \alpha > \log_5 2 \\ \frac{1}{2} & \text{if } \alpha = \log_5 2 \\ 0 & \text{if } \alpha < \log_5 2 \end{cases}$$

Exercise 3.32

Compute, as the parameter $\alpha \in \mathbb{R}$ varies, the limit of the sequence

$$\lim_{n\to+\infty} (e^{(2-\alpha)\,n} + 1) \log\left(1 + \frac{1}{e^n}\right)$$

Solution

We have

$$\begin{aligned} a_n = (e^{(2-\alpha)\,n} + 1) \log\left(1 + \frac{1}{e^n}\right) &= (e^{(2-\alpha)\,n} + 1) \frac{e^n}{e^n} \log\left(1 + \frac{1}{e^n}\right) \\ &= (e^{(2-\alpha)\,n} + 1) \frac{1}{e^n} \log\left(1 + \frac{1}{e^n}\right)^{e^n} \\ &\sim (e^{(2-\alpha)\,n} + 1) \frac{1}{e^n} \end{aligned}$$

since

$$\left(1+\frac{1}{e^n}\right)^{e^n} \to e \quad \text{and} \quad \log\left(1+\frac{1}{e^n}\right)^{e^n} \to 1$$

Hence

$$\lim_{n\to+\infty} a_n = \lim_{n\to+\infty} (e^{(2-\alpha)\,n}+1)\frac{1}{e^n}$$

On the other hand

$$(e^{(2-\alpha)\,n}+1)\frac{1}{e^n} = e^{(1-\alpha)n}+e^{-n} \to \begin{cases} +\infty & \text{if } 1-\alpha>0 \\ 1 & \text{if } 1-\alpha=0 \\ 0 & \text{if } 1-\alpha<0 \end{cases}$$

Summarizing, we obtain

$$\lim_{n\to+\infty} (e^{(2-\alpha)\,n}+1)\log\left(1+\frac{1}{e^n}\right) = \begin{cases} +\infty & \text{if } \alpha<1 \\ 1 & \text{if } \alpha=1 \\ 0 & \text{if } \alpha>1 \end{cases}$$

3.8 Multiple choice exercises

Exercise 3.33

Let

$$\lim_{n\to+\infty}\left(\frac{\sqrt{3}+n}{e-2n}\right)^n = \ell$$

(a) $\ell = 0$

(b) $\ell = e^{\left(\sqrt{3}-\frac{e}{2}\right)}$

(c) $\ell = +\infty$

(d) None of the other answers is correct.

Solution

Setting

$$a_n = \frac{\sqrt{3}+n}{e-2n}$$

we have $a_n < 0$ for $n \geq 2$ and

$$|a_n|^n = \left(\frac{\sqrt{3}+n}{2n-e}\right)^n \to 0$$

since $|a_n| \to 1/2$. From $|a_n|^n \to 0$ it follows that $(a_n)^n \to 0$. The correct answer is (a).

Exercise 3.34

Let

$$\lim_{n\to+\infty}(-1)^n n^\alpha\left(\frac{1+3n}{2-n}\right)^n=\ell$$

(a) ℓ does not exist if $\alpha\geq 0$

(b) $\ell=+\infty$ for every $\alpha\in\mathbb{R}$

(c) $\ell=0$ if $\alpha<0$

(d) None of the other answers is correct.

Solution

We set

$$a_n=(-1)^n\left(\frac{1+3n}{2-n}\right)^n=\left(\frac{1+3n}{n-2}\right)^n.$$

We have $a_n\to+\infty$ since $(1+3n)/(n-2)\to 3$. Let us compare a_n with 3^n:

$$\frac{a_n}{3^n}=\left(\frac{1+3n}{3n-6}\right)^n=\left[\left(1+\frac{1}{\frac{3n-6}{7}}\right)^{\frac{3n-6}{7}}\right]^{\frac{7n}{3n-6}}\to e^{7/3}$$

from which

$$a_n\sim 3^n e^{7/3}.$$

It follows that

$$\lim_{n\to+\infty}(-1)^n n^\alpha\left(\frac{1+3n}{2-n}\right)^n=\lim_{n\to+\infty}e^{7/3}n^\alpha 3^n=+\infty$$

even when $\alpha<0$, because 3^n is an infinity of higher order than any power of n. The correct answer is (b).

Exercise 3.35

Let

$$\lim_{n\to+\infty}\left(\frac{an+2}{en+a}\right)^n=\ell$$

(a) $\ell=e^{\left(\frac{2}{e}-1\right)}$ if $a=e$

(b) $\ell=1$ if $a\leq e$

(c) $\ell=0$ if $a>0$

(d) None of the other answers is correct.

Solution

Setting

$$a_n = \frac{an+2}{en+a}$$

we have $a_n \to a/e$, hence

$$(a_n)^n \to \begin{cases} +\infty, & a > e \\ 0, & 0 < a < e \end{cases}$$

therefore answers (b) and (c) are not correct. In the case $a = e$ we have an indeterminate form 1^∞:

$$\left(\frac{en+2}{en+e}\right)^n = \left[\left(1+\frac{1}{\frac{en+e}{2-e}}\right)^{\frac{en+e}{2-e}}\right]^{\frac{(2-e)n}{en+e}} \to e^{-1+\frac{2}{e}}.$$

The correct answer is (a).

Chapter 4

Additional exercises

Exercise 4.1

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty} \frac{2^{-n} + 3\,n^3}{\log^6 n + 2 - n^5}$$

Solution

Exercise 4.2

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty} \frac{\log n + 3\,n^3\,\log n}{2^{1/n} + n^5}$$

Solution

Exercise 4.3

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty} \frac{n^2 + \log^3 n}{n\,\log n + 1}$$

Solution

Exercise 4.4

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty} \frac{n^{2+\frac{1}{n}}}{1+n^{3-\frac{2}{n}}}$$

Solution

Exercise 4.5

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty} \left(\frac{n}{n+1}\right)^{2\,n}$$

Solution

Exercise 4.6

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty} \left(\frac{n+2}{n+1}\right)^{n^2}$$

Solution

Exercise 4.7

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty} \left(1+\frac{3}{n}\right)^{-n}$$

Solution

Exercise 4.8

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty}\left(1-\frac{2}{n}\right)^n$$

Solution

Exercise 4.9

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty}\frac{n+\log(n^2)-2^n}{(\log n)^3+n^2}$$

Solution

Exercise 4.10

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty}\frac{8+2\,n-n^{4/3}}{n^{-1/3}+3\,n-n^{5/3}}$$

Solution

Exercise 4.11

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty} n\left(\sqrt{n^2+3}-n\right)$$

Solution

Exercise 4.12

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty} \frac{n^2 + \log(5\,n^3) + (\log n)^2}{2\,(\log n)^3 + n\,\log n}$$

Solution

Exercise 4.13

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty} \frac{8 + 2\,n^2 - n^{4/3}}{n^{-1/3} + 3\,n^2 - n^{5/3}}$$

Solution

Exercise 4.14

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty} \left(\sqrt{n^4+1} - n\,\sqrt{n^2+1}\right)$$

Solution

Exercise 4.15

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty} \frac{n^2 + 3^n + 2^{-n}}{2^{2\,n} + (\log n)^3}$$

Solution

Exercise 4.16

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty} \frac{1}{n}\,(\sin^2 n)\cdot \sin(n^2)$$

Solution

Exercise 4.17

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n\to+\infty} \frac{\cos(\pi\, n)}{\pi\, n} + \log\left(\sin\frac{\pi}{n}\right)$$

Solution

Part II

Limits of functions and continuity

Chapter 1

Limits of real functions of a real variable

We have the following fundamental limit:

$$\left(1+\frac{\alpha}{\eta(x)}\right)^{\eta(x)} \to e^{\alpha} \qquad \text{as } \eta(x) \to \pm\infty, \quad \alpha \in \mathbb{R}, \alpha \neq 0$$

We have the following 5 main fundamental limits:

$$\frac{\sin \varepsilon(x)}{\varepsilon(x)} \to 1 \qquad \text{as } \varepsilon(x) \to 0$$

$$\frac{1-\cos \varepsilon(x)}{\varepsilon^2(x)} \to \frac{1}{2} \qquad \text{as } \varepsilon(x) \to 0$$

$$\frac{\log\big(1+\varepsilon(x)\big)}{\varepsilon(x)} \to 1 \qquad \text{as } \varepsilon(x) \to 0$$

$$\frac{e^{\varepsilon(x)}-1}{\varepsilon(x)} \to 1 \qquad \text{as } \varepsilon(x) \to 0$$

$$\frac{\big(1+\varepsilon(x)\big)^{\alpha}-1}{\varepsilon(x)} \to \alpha \qquad \text{as } \varepsilon(x) \to 0, \quad \alpha \in \mathbb{R}, \alpha \neq 0$$

We also have the following fundamental limits of the hyperbolic functions:

$$\frac{\sinh \varepsilon(x)}{\varepsilon(x)} \to 1 \qquad \text{as } \varepsilon(x) \to 0$$

$$\frac{\cosh \varepsilon(x)-1}{\varepsilon^2(x)} \to \frac{1}{2} \qquad \text{as } \varepsilon(x) \to 0$$

By the hierarchy of infinities theorem, we have:

$$\varepsilon^{\beta}(x)\,\big[\log_a \varepsilon(x)\big]^{\alpha} \to 0 \qquad \text{as } \varepsilon(x) \to 0^{+}, \quad \alpha, \beta > 0, \quad a > 1$$

From the 5 main fundamental limits, the following asymptotic equivalences follow:

$$\sin \varepsilon(x) \sim \varepsilon(x) \qquad \text{as } \varepsilon(x) \to 0$$

$$1 - \cos \varepsilon(x) \sim \frac{1}{2}\,\varepsilon^2(x) \qquad \text{as } \varepsilon(x) \to 0$$

$$e^{\varepsilon(x)} - 1 \sim \varepsilon(x) \qquad \text{as } \varepsilon(x) \to 0$$

$$\log\big(1 + \varepsilon(x)\big) \sim \varepsilon(x) \qquad \text{as } \varepsilon(x) \to 0$$

$$\big(1 + \varepsilon(x)\big)^{\alpha} - 1 \sim \alpha\,\varepsilon(x) \qquad \text{as } \varepsilon(x) \to 0$$

From the 5 main fundamental limits, the following asymptotic expansions follow:

$$\sin \varepsilon(x) = \varepsilon(x) + o\big(\varepsilon(x)\big) \qquad \text{as } \varepsilon(x) \to 0$$

$$\cos \varepsilon(x) = 1 - \frac{1}{2}\,\varepsilon^2(x) + o\big(\varepsilon^2(x)\big) \qquad \text{as } \varepsilon(x) \to 0$$

$$\log\big(1 + \varepsilon(x)\big) = \varepsilon(x) + o\big(\varepsilon(x)\big) \qquad \text{as } \varepsilon(x) \to 0$$

$$e^{\varepsilon(x)} = 1 + \varepsilon(x) + o\big(\varepsilon(x)\big) \qquad \text{as } \varepsilon(x) \to 0$$

$$\big(1 + \varepsilon(x)\big)^{\alpha} = 1 + \alpha\,\varepsilon(x) + o\big(\varepsilon(x)\big) \qquad \text{as } \varepsilon(x) \to 0$$

From the fundamental limits, the following asymptotic equivalences also follow:

$$\sinh \varepsilon(x) \sim \varepsilon(x) \qquad \text{as } \varepsilon(x) \to 0$$

$$\cosh \varepsilon(x) - 1 \sim \frac{1}{2}\,\varepsilon^2(x) \qquad \text{as } \varepsilon(x) \to 0$$

For the hyperbolic functions we have the asymptotic expansions:

$$\sinh \varepsilon(x) = \varepsilon(x) + o\big(\varepsilon(x)\big) \qquad \text{as } \varepsilon(x) \to 0$$

$$\cosh \varepsilon(x) = 1 + \frac{1}{2}\,\varepsilon^2(x) + o\big(\varepsilon^2(x)\big) \qquad \text{as } \varepsilon(x) \to 0$$

Operations with "little o":

$o(1)$ denotes a generic infinitesimal;

$C \cdot o(x^n) = o(x^n)$;

$x^m \cdot o(x^n) = o(x^{m+n})$;

$o(x^m) \cdot o(x^n) = o(x^{m+n})$

$\frac{o(x^n)}{x^m} = o(x^{n-m}) \quad (n \geq m)$;

$o(x^n) \pm o(x^m) = o(x^M) \qquad$ where $M = \min\{m, n\}$.

Exercise 1.1

From $e^x = 1 + x + o(x)$, $x \to 0$, deduce that

$$\sinh(x) = x + o(x), \ x \to 0.$$

Solution

For $x \to 0$, we have $e^{-x} = 1 - x + o(x)$, hence

$$\sinh x = \frac{e^x - e^{-x}}{2} = \frac{1}{2}\Big(\big(1 + x + o(x)\big) - \big(1 - x + o(x)\big)\Big) = x + o(x).$$

Exercise 1.2

Using $\cosh^2 x - 1 = \sinh^2 x$ and the previous exercise, prove that

$$\lim_{x\to 0} \frac{\cosh x - 1}{x^2} = \frac{1}{2}$$

and deduce

$$\cosh x = 1 + \frac{1}{2}x^2 + o(x^2), \ x \to 0.$$

Solution

From the previous exercise we have $\sinh^2 x \sim x^2$ as $x \to 0$ and moreover $\cosh 0 = 1$, hence

$$\lim_{x\to 0} \frac{\cosh x - 1}{x^2} \cdot \frac{\cosh x + 1}{\cosh x + 1} = \lim_{x\to 0} \frac{\cosh^2 x - 1}{x^2} \cdot \underbrace{\lim_{x\to 0} \frac{1}{\cosh x + 1}}_{\to \frac{1}{2}} =$$

$$= \frac{1}{2} \lim_{x\to 0} \frac{\sinh^2 x}{x^2} = \frac{1}{2} \lim_{x\to 0} \frac{x^2}{x^2} = \frac{1}{2}.$$

Moreover

$$\left(\lim_{x\to 0}\frac{\cosh x-1}{x^2}\right)-\frac{1}{2}=\lim_{x\to 0}\frac{\cosh x-1-\frac{1}{2}x^2}{x^2}=0$$

which means

$$\cosh x-1-\frac{1}{2}x^2=o(x^2),$$

$$\cosh x=1+\frac{1}{2}x^2+o(x^2),\quad x\to 0.$$

Exercise 1.3

Refine the equivalence

$$\log(x^2+1)\sim 2\log x,\ x\to+\infty$$

by proving that

$$\log(x^2+1)=2\log x+\frac{1}{x^2}+o\left(\frac{1}{x^2}\right),\ x\to+\infty.$$

Solution

$$\log(x^2+1)=\log\left(x^2\left(1+\frac{1}{x^2}\right)\right)=\log x^2+\log\left(1+\frac{1}{x^2}\right).$$

Now

$$1/x^2\to 0\ \text{ as }\ x\to+\infty$$

and

$$\log(1+y)=y+o(y)\ \text{ as }\ y\to 0,$$

hence

$$\log(x^2+1)=2\log x+\frac{1}{x^2}+o\left(\frac{1}{x^2}\right),\ x\to+\infty.$$

Exercise 1.4

Compute the following limit of a function of a real variable:

$$\lim_{x\to+\infty}\log x-\sqrt{x}$$

Solution

$$\lim_{x\to+\infty}\log x-\sqrt{x}=\lim_{x\to+\infty}-\sqrt{x}=-\infty.$$

Exercise 1.5

Compute the following limit of a function of a real variable:

$$\lim_{x\to+\infty} 2^x - x^2$$

Solution

$$\lim_{x\to+\infty} 2^x - x^2 = \lim_{x\to+\infty} 2^x = +\infty.$$

Exercise 1.6

Compute the following limit of a function of a real variable:

$$\lim_{x\to+\infty} \frac{\log(x^2+1)}{2^x}$$

Solution

We use the equivalence $\log(x^2+1) \sim 2\log x$ as $x \to +\infty$:

$$\lim_{x\to+\infty} \frac{\log(x^2+1)}{2^x} = \lim_{x\to+\infty} \frac{2\log x}{2^x} = 0.$$

Exercise 1.7

Compute the following limit of a function of a real variable:

$$\lim_{x\to+\infty} \left(\frac{x+2}{x+1}\right)^x$$

Solution

We use the fundamental limit

$$\lim_{x\to\pm\infty} \left(1+\frac{1}{x}\right)^x = e.$$

$$\lim_{x\to+\infty} \left(\frac{x+2}{x+1}\right)^x = \lim_{x\to+\infty} \left[\left(1+\frac{1}{x+1}\right)^{x+1}\right]^{\frac{x}{x+1}} = e^1 = e.$$

Exercise 1.8

Compute the following limit of a function of a real variable:

$$\lim_{x\to 0^+} x^{\log x}$$

Solution

We have

$$x^{\log x} = e^{\log x^{\log x}} = e^{\log x\ \log x},$$

hence

$$\lim_{x\to 0^+} x^{\log x} = \lim_{x\to 0^+} e^{\overbrace{\log x}^{\to -\infty}\ \overbrace{\log x}^{\to -\infty}} = e^{+\infty} = +\infty.$$

Moreover, the base x tends to 0 and the exponent $\log x$ tends to $-\infty$, hence the limit is not an indeterminate form:

$$\lim_{x\to 0^+} x^{\log x} = 0^{-\infty} = +\infty$$

Exercise 1.9

Let

$$\lim_{x\to+\infty} x\left(\log(x+2) - \log x\right) = L.$$

(a) $L = 2$

(b) L does not exist

(c) $L = 1$

(d) None of the other answers is correct.

Solution

$$\lim_{x\to+\infty} x\left(\log(x+2) - \log x\right) = \lim_{x\to+\infty} x \log\left(1 + \frac{2}{x}\right) = \lim_{x\to+\infty} x \cdot \frac{2}{x} = 2.$$

The correct answer is (a).

Exercise 1.10

Compute the following limit of a quotient using equivalent power functions for each term:

$$\lim_{x\to 0}\frac{e^x-1}{\sin x}$$

Solution

$$\lim_{x\to 0}\frac{e^x-1}{\sin x}=\lim_{x\to 0}\frac{x}{x}=1.$$

Exercise 1.11

Compute the following limit of a quotient using equivalent power functions for each term:

$$\lim_{x\to 0}\frac{1-\cos x}{\sin^2 x}$$

Solution

We have

$$\lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2}\quad\text{hence}\quad 1-\cos x\sim\frac{1}{2}\,x^2\quad\text{as}\quad x\to 0.$$

Moreover, we have

$$\sin x\sim x\quad\text{as}\quad x\to 0\quad\text{hence}\quad(\sin x)^2\sim x^2\quad\text{as}\quad x\to 0.$$

We then have:

$$\lim_{x\to 0}\frac{1-\cos x}{\sin^2 x}=\lim_{x\to 0}\frac{\frac{1}{2}x^2}{x^2}=\frac{1}{2}$$

Exercise 1.12

Compute the following limit of a quotient using equivalent power functions for each term:

$$\lim_{x\to 0}\frac{\sin 3x}{\sin 4x}$$

Solution

$$\lim_{x\to 0}\frac{\sin 3x}{\sin 4x}=\lim_{x\to 0}\frac{3x}{4x}=\frac{3}{4}$$

Exercise 1.13

Compute the following limit of a quotient using equivalent power functions for each term:

$$\lim_{x\to 0} \frac{(1-e^{2x})^2}{1-\cos 5x}$$

Solution

$$\lim_{x\to 0} \frac{(1-e^{2x})^2}{1-\cos 5x} = \lim_{x\to 0} \frac{(-2x)^2}{\frac{1}{2}(5x)^2} = \frac{8}{25}$$

Exercise 1.14

Compute the following limit of a quotient using equivalent power functions for each term:

$$\lim_{x\to 0} \frac{\sin x^3}{(1-e^x)^3}$$

Solution

$$\lim_{x\to 0} \frac{\sin x^3}{(1-e^x)^3} = \lim_{x\to 0} \frac{x^3}{(-x)^3} = -1.$$

Exercise 1.15

Compute

$$\lim_{x\to 0} x^2 e^{\frac{\sqrt{\pi}}{x}\sin(x\log 7)}$$

Solution

We analyze the exponent using the equivalence $\sin(x\log 7) \sim x\log 7$ as $x \to 0$:

$$\lim_{x\to 0} \frac{\sqrt{\pi}}{x}\sin(x\log 7) = \lim_{x\to 0} \frac{\sqrt{\pi}}{x}\cdot x\log 7 = \sqrt{\pi}\log 7.$$

The given limit equals

$$\lim_{x\to 0} x^2 e^{\frac{\sqrt{\pi}}{x}\sin(x\log 7)} = 0\cdot e^{\sqrt{\pi}\log 7} = 0\cdot 7^{\sqrt{\pi}} = 0.$$

Exercise 1.16

Compute

$$\lim_{x\to 0^+}\left(\frac{1}{\sin x}+\log x\right)$$

Solution

Using

$$\lim_{x\to 0^+}\sin x\log x=\lim_{x\to 0^+}x\log x=0,$$

we have

$$\lim_{x\to 0^+}\left(\frac{1}{\sin x}+\log x\right)=\lim_{x\to 0^+}\frac{1+\sin x\log x}{\sin x}=+\infty$$

since the numerator $1+\sin x\log x$ tends to 1 while the denominator $\sin x$ is a positive function tending to 0.

Equivalently, again from

$$\lim_{x\to 0^+}\frac{\log x}{\frac{1}{\sin x}}=\lim_{x\to 0^+}\sin x\log x=\lim_{x\to 0^+}x\log x=0,$$

we have that $1/\sin x$ is an infinity of higher order than $\log x$ as $x\to 0^+$, hence

$$\lim_{x\to 0^+}\left(\frac{1}{\sin x}+\log x\right)=\lim_{x\to 0^+}\frac{1}{\sin x}=+\infty.$$

We also prove that

$$\lim_{x\to 0^+}x\ \log x=0$$

We make the change of variable $x=\frac{1}{y}$, hence

$$\lim_{x\to 0^+}x\ \log x=\lim_{y\to+\infty}\frac{1}{y}\ \log\frac{1}{y}=\lim_{y\to+\infty}\frac{-\log y}{y}=0$$

Exercise 1.17

Compute

$$\lim_{x\to 0}\frac{(1-\cos x)^3-\sin x^6}{x^6}$$

Solution

From

$$1-\cos x\sim\frac{1}{2}x^2,\ \sin x\sim x,\quad x\to 0,$$

we have

$$\lim_{x\to 0}\frac{(1-\cos x)^3-\sin x^6}{x^6}=\lim_{x\to 0}\frac{(1-\cos x)^3}{x^6}-\lim_{x\to 0}\frac{\sin x^6}{x^6}=$$

$$=\lim_{x\to 0}\frac{\frac{1}{8}x^6}{x^6}-\lim_{x\to 0}\frac{x^6}{x^6}=\frac{1}{8}-1=-\frac{7}{8}.$$

Equivalently, using the calculus of Landau symbols, from

$$\cos x=1-\frac{1}{2}x^2+o(x^2),\ \sin x=x+o(x),\quad x\to 0,$$

we have

$$\lim_{x\to 0}\frac{(1-\cos x)^3-\sin x^6}{x^6}=\lim_{x\to 0}\frac{\frac{1}{8}x^6-x^6+o(x^6)}{x^6}=$$

$$\lim_{x\to 0}\frac{-\frac{7}{8}x^6+o(x^6)}{x^6}=\lim_{x\to 0}\frac{-\frac{7}{8}x^6}{x^6}=-\frac{7}{8}.$$

Exercise 1.18

Compute

$$\lim_{x\to 0}\frac{8(1-\cos x)^3-\sin x^6}{x^6}$$

Solution

From

$$1-\cos x\sim\frac{1}{2}x^2,\ \sin x\sim x,\quad x\to 0,$$

we have

$$\lim_{x\to 0}\frac{8(1-\cos x)^3-\sin x^6}{x^6}=\lim_{x\to 0}\frac{8(1-\cos x)^3}{x^6}-\lim_{x\to 0}\frac{\sin x^6}{x^6}=$$

$$=\lim_{x\to 0}\frac{x^6}{x^6}-\lim_{x\to 0}\frac{x^6}{x^6}=1-1=0.$$

Equivalently, using the calculus of Landau symbols, from

$$\cos x=1-\frac{1}{2}x^2+o(x^2),\ \sin x=x+o(x),\quad x\to 0,$$

we have

$$\lim_{x\to 0}\frac{8(1-\cos x)^3-\sin x^6}{x^6}=\lim_{x\to 0}\frac{x^6-x^6+o(x^6)}{x^6}=$$

$$\lim_{x\to 0}\frac{o(x^6)}{x^6}=0$$

since the numerator $o(x^6)$, by the very definition of the Landau symbol, is an infinitesimal of higher order than the denominator x^6.

We have the asymptotic expansions as $x\to 0$:

$$\sin x = x + o(x),\ \cos x = 1 - \tfrac{1}{2}x^2 + o(x^2),\ e^x = 1 + x + o(x),$$

$$\sinh x = x + o(x),\ \cosh x = 1 + \tfrac{1}{2}x^2 + o(x^2).$$

Exercise 1.19

Compute the following limits using the comparison of infinitesimals

$$\lim_{x\to 0} \frac{(\sin x)^2 + x}{x^3 - \sin x}$$

Solution

$\lim_{x\to0} \frac{(\sin x)^2+x}{x^3-\sin x} = \lim_{x\to0} \frac{(x+o(x))^2+x}{x^3-x+o(x)} =$

$= \lim_{x\to0} \frac{x^2+o(x^2)+x}{x^3-x+o(x)} = \lim_{x\to0} \frac{x+o(x)}{-x+o(x)} = \lim_{x\to0} \frac{x}{-x} = -1.$

Exercise 1.20

Compute the following limits using the comparison of infinitesimals

$$\lim_{x\to 0} \frac{|1 - \cos x + \sin x|}{(e^x - 1)^2}$$

Solution

$\lim_{x\to0} \frac{|1-\cos x+\sin x|}{(e^x-1)^2} = \lim_{x\to0} \frac{|\frac{1}{2}x^2+o(x^2)+x+o(x)|}{(x+o(x))^2} =$

$\lim_{x\to0} \frac{|x+o(x)|}{x^2+o(x^2)} = \lim_{x\to0} \frac{|x|}{x^2} = \lim_{x\to0} \frac{1}{|x|} = +\infty.$

Exercise 1.21

Compute the following limits using the comparison of infinitesimals

$$\lim_{x\to 0} \frac{\sin 2x - \sin^2 x}{e^x - 1}$$

Solution

$$\lim_{x\to 0} \frac{\sin 2x - \sin^2 x}{e^x - 1} = \lim_{x\to 0} \frac{2x + o(x) - (x + o(x))^2}{x + o(x)} =$$

$$= \lim_{x\to 0} \frac{2x + o(x) - x^2 + o(x^2)}{x + o(x)} = \lim_{x\to 0} \frac{2x + o(x)}{x + o(x)} = \lim_{x\to 0} \frac{2x}{x} = 2.$$

Exercise 1.22

Compute the following limits using the comparison of infinitesimals

$$\lim_{x\to 0} \frac{\sin^3 x - \sin 4x}{\cosh x - 1 + x}$$

Solution

$$\lim_{x\to 0} \frac{\sin^3 x - \sin 4x}{\cosh x - 1 + x} = \lim_{x\to 0} \frac{(x + o(x))^3 - 4x + o(x)}{\frac{1}{2}x^2 + o(x^2) + x} =$$

$$= \lim_{x\to 0} \frac{x^3 + o(x^3) - 4x + o(x)}{x + o(x)} = \lim_{x\to 0} \frac{-4x + o(x)}{x + o(x)} = \lim_{x\to 0} \frac{-4x}{x} = -4.$$

Exercise 1.23

Compute the following limits using the comparison of infinitesimals

$$\lim_{x\to 0} \frac{\sin x - \sin x^2}{\sinh(2x) - \sinh(2x)^2}$$

Solution

$$\lim_{x\to 0} \frac{\sin x - \sin x^2}{\sinh(2x) - \sinh(2x)^2} = \lim_{x\to 0} \frac{x + o(x) - x^2 + o(x^2)}{2x + o(x) - (2x + o(x))^2} =$$

$$= \lim_{x\to 0} \frac{x + o(x)}{2x + o(x) - 4x^2 + o(x^2)} = \lim_{x\to 0} \frac{x + o(x)}{2x + o(x)} = \lim_{x\to 0} \frac{x}{2x} = \frac{1}{2}.$$

Exercise 1.24

Compute

$$\lim_{x\to 1^+} \frac{\log(1 + \sqrt{x - 1})}{\sqrt{x^2 - 1}}$$

Solution

We have

$$\lim_{x\to 1^+} \frac{\log(1+\sqrt{x-1})}{\sqrt{x^2-1}} = \lim_{x\to 1^+} \frac{\log(1+\sqrt{x-1})}{\sqrt{x-1}\sqrt{x+1}} = \frac{1}{\sqrt{2}} \lim_{x\to 1^+} \frac{\log(1+\sqrt{x-1})}{\sqrt{x-1}}.$$

Setting $y = \sqrt{x-1}$, the limit becomes

$$\frac{1}{\sqrt{2}} \lim_{y\to 0^+} \frac{\log(1+y)}{y} = \frac{1}{\sqrt{2}}$$

by the fundamental limit

$$\lim_{y\to 0} \frac{\log(1+y)}{y} = 1.$$

Exercise 1.25

Compute

$$\lim_{x\to+\infty} \left(\frac{x^2+3x}{x^2-5x}\right)^{(x+\log x)}$$

Solution

We have

$$\lim_{x\to+\infty} \left(\frac{x^2+3x}{x^2-5x}\right)^{(x+\log x)} = \lim_{x\to+\infty} \left(\frac{x+3}{x-5}\right)^{(x+\log x)} = \lim_{x\to+\infty} \left(1+\frac{8}{x-5}\right)^{(x+\log x)}$$
$$= \lim_{x\to+\infty} \left(1+\frac{1}{\frac{x-5}{8}}\right)^{(x+\log x)}$$

Using the fundamental limit

$$\lim_{x\to\pm\infty} \left(1+\frac{1}{x}\right)^{x} = e$$

we obtain

$$\lim_{x\to+\infty} \left(\frac{x^2+3x}{x^2-5x}\right)^{(x+\log x)} = \lim_{x\to+\infty} \left[\left(1+\frac{1}{\frac{x-5}{8}}\right)^{\frac{x-5}{8}}\right]^{\frac{8(x+\log x)}{x-5}} = e^8$$

since

$$\frac{8(x+\log x)}{x-5} \sim \frac{8x}{x-5} \xrightarrow{x\to+\infty} 8.$$

Exercise 1.26

Compute

$$\lim_{x\to 0} \frac{x\left(\pi^x - e^x\right)}{\cos(x)-1}$$

Solution

We have

$$\lim_{x\to 0}\frac{x\left(\pi^x-e^x\right)}{\cos(x)-1}=\lim_{x\to 0}\frac{x\pi^x\left(1-\left(\frac{e}{\pi}\right)^x\right)}{\cos(x)-1}=\lim_{x\to 0}\pi^x\frac{x\left(1-\left(\frac{e}{\pi}\right)^x\right)}{\cos(x)-1}$$

Since

$$1-\left(\frac{e}{\pi}\right)^x\sim -x\log\left(\frac{e}{\pi}\right)$$

and also

$$\cos(x)-1\sim -\frac{1}{2}x^2$$

we obtain

$$\lim_{x\to 0}\frac{x\left(\pi^x-e^x\right)}{\cos(x)-1}=\lim_{x\to 0}\pi^x\frac{-x^2\log\left(\frac{e}{\pi}\right)}{-\frac{1}{2}x^2}=2\log\left(\frac{e}{\pi}\right).$$

Exercise 1.27

Compute

$$\lim_{x\to 0}\frac{\sin\left(\ln\left(1-x\right)\right)}{1-2^x}$$

Solution

The limit is in fact immediate if we observe that, as $x\to 0$,

$$\ln\left(1-x\right)\sim -x$$

and also

$$1-2^x\sim -x\log 2$$

thus obtaining

$$\lim_{x\to 0}\frac{\sin\left(\ln\left(1-x\right)\right)}{1-2^x}=\lim_{x\to 0}\frac{\sin(-x)}{-x\log 2}=\frac{1}{\log 2}$$

since

$$\sin(-x)\sim -x$$

Exercise 1.28

Compute

$$\lim_{x\to+\infty}\frac{x^4\sin^2(\pi-2\arctan(x))}{x^2+3}$$

Solution

Recalling that

$$\arctan(x)\xrightarrow{x\to+\infty}\frac{\pi}{2}$$

and hence that

$$\pi-2\arctan(x)\xrightarrow{x\to+\infty}0$$

we can immediately use the asymptotic estimate

$$\sin^2(\pi - 2\arctan(x)) \sim (\pi - 2\arctan(x))^2$$

as well as

$$\frac{x^4}{x^2+3} \sim x^2$$

and say that

$$\begin{aligned}\lim_{x\to+\infty} \frac{x^4 \sin^2(\pi - 2\arctan(x))}{x^2+3} &= \lim_{x\to+\infty} \frac{x^4(\pi - 2\arctan(x))^2}{x^2+3} \\ &= \lim_{x\to+\infty} x^2(\pi - 2\arctan(x))^2\end{aligned}$$

which is still an indeterminate form of the type $[0 \cdot +\infty]$. However, with the change of variable $y = \pi - 2\arctan(x)$ and observing that

$$x = \tan\left(\frac{\pi}{2} - \frac{y}{2}\right) = \cot\left(\frac{y}{2}\right)$$

we obtain

$$\begin{aligned}\lim_{x\to+\infty} x^2 \left(\pi - 2\arctan(x)\right)^2 &= \lim_{y\to 0} y^2 \cot^2\left(\frac{y}{2}\right) \\ &= \lim_{y\to 0} \frac{y^2}{\sin^2\left(\frac{y}{2}\right)} \cos^2\left(\frac{y}{2}\right) = 4.\end{aligned}$$

Exercise 1.29

Compute

$$\lim_{x\to 1} \frac{\log\left(x^x\right) - \log x}{1 - \cos(x-1)}$$

Solution

Recalling that

$$\log\left(x^x\right) - \log x = \log(e^{x\log x}) - \log x = x\log x - \log x = (x-1)\log x$$

we immediately have

$$\lim_{x\to 1} \frac{\log\left(x^x\right) - \log x}{1 - \cos(x-1)} = \lim_{x\to 1} \frac{(x-1)\log x}{1 - \cos(x-1)}$$

Setting, for greater clarity in the use of the asymptotic estimates, $x - 1 = t$, we obtain by substitution

$$\lim_{x\to 1} \frac{(x-1)\log x}{1 - \cos(x-1)} = \lim_{t\to 0} \frac{t\log(1+t)}{1-\cos t} = 2$$

since

$$\frac{t\log(1+t)}{1-\cos t} \sim \frac{t^2}{\frac{1}{2}t^2}$$

Exercise 1.30

Compute

$$\lim_{x\to+\infty} x \log\left(\frac{3x+x^2}{1+x+x^2}\right)$$

Solution

We immediately observe that the limit is in the indeterminate form of the type $[+\infty \cdot 0]$, since

$$\frac{3x+x^2}{1+x+x^2} \xrightarrow{x\to+\infty} 1$$

In order to resolve the indeterminate form, we can add and subtract 1 inside the argument of the logarithm:

$$\log\left(\frac{3x+x^2}{1+x+x^2}\right) = \log\left(1+\frac{3x+x^2}{1+x+x^2}-1\right)$$

At this point, since

$$\frac{3x+x^2}{1+x+x^2}-1 \xrightarrow{x\to+\infty} 0$$

we can use the well-known asymptotic estimate of the logarithm, valid for some $\varepsilon(x) \to 0$, obtaining

$$\log\left(1+\frac{3x+x^2}{1+x+x^2}-1\right) \sim \frac{3x+x^2}{1+x+x^2}-1 = \frac{2x-1}{1+x+x^2}$$

The limit reduces to

$$\lim_{x\to+\infty} x\left(\frac{2x-1}{1+x+x^2}\right) = 2.$$

Exercise 1.31

Compute

$$\lim_{x\to+\infty} \left(e^{\sqrt{x^2+x}} - e^{\sqrt{x^2-1}}\right)$$

Solution

Factoring out $e^{\sqrt{x^2-1}}$ we obtain

$$\lim_{x\to+\infty} e^{\sqrt{x^2-1}}\left(e^{\sqrt{x^2+x}-\sqrt{x^2-1}}-1\right).$$

We examine the indeterminate form $\sqrt{x^2+x}-\sqrt{x^2-1}$ as $x\to+\infty$:

$$\lim_{x\to+\infty} \sqrt{x^2+x}-\sqrt{x^2-1} \cdot \frac{\sqrt{x^2+x}+\sqrt{x^2-1}}{\sqrt{x^2+x}+\sqrt{x^2-1}} = \lim_{x\to+\infty} \frac{x^2+x-x^2+1}{\sqrt{x^2+x}+\sqrt{x^2-1}}$$

$$= \lim_{x\to+\infty} \frac{x+1}{x\left(\sqrt{1+\frac{1}{x}}+\sqrt{1-\frac{1}{x^2}}\right)} = \frac{1}{2}.$$

Hence the given limit equals

$$\lim_{x\to+\infty} e^{\sqrt{x^2-1}}\left(e^{\sqrt{x^2+x}-\sqrt{x^2-1}}-1\right) = (\sqrt{e}-1)\lim_{x\to+\infty} e^{\sqrt{x^2-1}} = +\infty.$$

Equivalently, from $1/x \to 0$ as $x \to +\infty$ and from

$$\sqrt{1+y} = 1 + (1/2)y + o(y)$$

as $y \to 0$, we have

$$\sqrt{x^2+x} = x\sqrt{1+\frac{1}{x}} = x\left(1+\frac{1}{2x}+o\left(\frac{1}{x}\right)\right) = x+\frac{1}{2}+o(1), \quad x\to+\infty,$$

where $o(1)$ denotes a generic infinitesimal, and

$$\sqrt{x^2-1} = x\sqrt{1-\frac{1}{x^2}} = x\left(1-\frac{1}{2x^2}+o\left(\frac{1}{x^2}\right)\right) = x-\frac{1}{2x}+o\left(\frac{1}{x}\right), \quad x\to+\infty.$$

Hence the given limit equals

$$\lim_{x\to+\infty}\left(e^{\sqrt{x^2+x}}-e^{\sqrt{x^2-1}}\right) = \lim_{x\to+\infty}\left(e^{x+1/2+o(1)}-e^{x-1/(2x)+o(1/x)}\right) =$$

$$\lim_{x\to+\infty} e^x\left(e^{1/2+o(1)}-e^{-1/(2x)+o(1/x)}\right) = (\sqrt{e}-1)\lim_{x\to+\infty} e^x = +\infty.$$

We have that $1/x \to 0$ as $x \to +\infty$ and we have the asymptotic expansions as $y \to 0$

$$\sin y = y+o(y),\ \cos y = 1-\frac{1}{2}y^2+o(y^2),\ e^y = 1+y+o(y),\ \log(1+y) = y+o(y).$$

Exercise 1.32

Compute the following limit

$$\lim_{x\to+\infty}\frac{1+x^4\sin(1/x^4)}{x^2(1-\cos(1/x^2))}$$

Solution

From the asymptotic expansion we have:

$$\cos\left(\frac{1}{x^2}\right) = 1-\frac{1}{2}\left(\frac{1}{x^2}\right)^2+o\left(\left(\frac{1}{x^2}\right)^2\right) = 1-\frac{1}{2}\frac{1}{x^4}+o\left(\frac{1}{x^4}\right) \quad \text{as} \quad x\to+\infty.$$

Expanding the ratio asymptotically, we have:

- for the numerator

$$1 + x^4 \sin\left(\frac{1}{x^4}\right) = 1 + x^4\left(\frac{1}{x^4} + o\left(\frac{1}{x^4}\right)\right) = 1 + 1 + o(1) \quad \text{as} \quad x \to +\infty;$$

- for the denominator

$$x^2\left[1 - \cos\left(\frac{1}{x^2}\right)\right] = x^2\left[1 - \left(1 - \frac{1}{2}\,\frac{1}{x^4} + o\left(\frac{1}{x^4}\right)\right)\right] =$$

$$= x^2\left[\frac{1}{2}\,\frac{1}{x^4} - o\left(\frac{1}{x^4}\right)\right] = \frac{1}{2}\,\frac{1}{x^2} - o\left(\frac{1}{x^2}\right) = \frac{1}{2}\,\frac{1}{x^2} + o\left(\frac{1}{x^2}\right) \quad \text{as} \quad x \to +\infty.$$

Hence we obtain:

$$\lim_{x\to+\infty} \frac{1 + x^4 \sin\left(\frac{1}{x^4}\right)}{x^2\left[1 - \cos\left(\frac{1}{x^2}\right)\right]} = \lim_{x\to+\infty} \frac{2 + o(1)}{\frac{1}{2}\,\frac{1}{x^2} + o\left(\frac{1}{x^2}\right)} = +\infty$$

where, as always, $o(1)$ denotes a generic infinitesimal.

Exercise 1.33

Compute the following limit

$$\lim_{x\to+\infty} \frac{x^2 e^{-1/x^4} - x^2}{\log(x^2+1) - 2\log x}$$

Solution

$$\lim_{x\to+\infty} \frac{x^2\left(e^{-1/x^4}-1\right)}{\log(x^2+1)-\log x^2} = \lim_{x\to+\infty} \frac{x^2\left(-\frac{1}{x^4}+o\left(\frac{1}{x^4}\right)\right)}{\log\left(1+\frac{1}{x^2}\right)} =$$

$$= \lim_{x\to+\infty} \frac{-\frac{1}{x^2}+o\left(\frac{1}{x^2}\right)}{\frac{1}{x^2}+o\left(\frac{1}{x^2}\right)} = \lim_{x\to+\infty} \frac{-\frac{1}{x^2}+o\left(\frac{1}{x^2}\right)}{\frac{1}{x^2}+o\left(\frac{1}{x^2}\right)} = \lim_{x\to+\infty} \frac{-\frac{1}{x^2}}{\frac{1}{x^2}} = -1.$$

Exercise 1.34

Compute

$$\lim_{x\to0} \frac{e^{-x^4} - x^4 - 1}{x^2\cos x - x^2}$$

Solution

We use the expansion

$$e^{-x^4} = 1 - x^4 + o(x^4) \quad \text{as} \quad x \to 0$$

and the equivalence

$$\cos x - 1 \sim -x^2/2 \quad \text{as} \quad x \to 0.$$

Hence we have

$$\lim_{x\to 0} \frac{e^{-x^4} - x^4 - 1}{x^2(\cos x - 1)} = \lim_{x\to 0} \frac{1 - x^4 + o(x^4) - x^4 - 1}{x^2\left(-\frac{x^2}{2}\right)} =$$

$$= \lim_{x\to 0} \frac{-2x^4 + o(x^4)}{-\frac{x^4}{2}} = \lim_{x\to 0} \frac{-2x^4}{-\frac{x^4}{2}} = 4.$$

Exercise 1.35

Compute

$$\lim_{x\to \pi/4} (2\sin^2 x)^{\frac{1}{\cos 2x}}$$

Solution

Using $\cos 2x = 1 - 2\sin^2 x$ and setting $y = 1 - 2\sin^2 x$, we observe that $y \to 0$ as $x \to \pi/4$ and the given limit equals

$$\lim_{y\to 0} (1-y)^{\frac{1}{y}} = \lim_{y\to 0} e^{\frac{\log(1-y)}{y}} = e^{-1} = \frac{1}{e}$$

also by virtue of the fundamental limit

$$\lim_{y\to 0} \frac{\log(1-y)}{y} = -1.$$

Exercise 1.36

Compute

$$\lim_{x\to 0^+} \frac{(1+x^5)^{\frac{1}{x^2 \sin 2x}} - 1}{2\log(1+x^3)}$$

Solution

We write the numerator in the form

$$e^{\frac{\log(1+x^5)}{x^2 \sin 2x}} - 1$$

and we analyze the exponent. As $x \to 0$ we have

$$\frac{\log(1+x^5)}{x^2 \sin 2x} \sim \frac{x^5}{x^2 \cdot 2x} = \frac{x^2}{2}.$$

From this and from $e^y - 1 \sim y$ as $y \to 0$, it follows that

$$e^{\frac{\log(1+x^5)}{x^2 \sin 2x}} - 1 \sim \frac{\log(1+x^5)}{x^2 \sin 2x} \sim \frac{x^2}{2}.$$

The given limit equals

$$\lim_{x\to 0^+} \frac{(1+x^5)^{\frac{1}{x^2\sin 2x}}-1}{2\log(1+x^3)} = \lim_{x\to 0^+} \frac{\frac{x^2}{2}}{2x^3} = \frac{1}{4}\lim_{x\to 0^+}\frac{1}{x} = +\infty.$$

Exercise 1.37

Compute

$$\lim_{x\to 0^+} \frac{1-x^x}{x^2}$$

Solution

Writing the numerator in the form

$$1-e^{x\log x},$$

the exponent $x\log x$ tends to zero as $x\to 0^+$. Since $1-e^y\sim -y$ as $y\to 0$, it follows that

$$1-e^{x\log x}\sim -x\log x, \quad x\to 0^+.$$

The given limit equals

$$\lim_{x\to 0^+} \frac{1-x^x}{x^2} = \lim_{x\to 0^+} \frac{-x\log x}{x^2} = \lim_{x\to 0^+} \frac{-\log x}{x} = +\infty.$$

Exercise 1.38

Compute

$$\lim_{x\to 0} \frac{\sin(e^{x^3}-1)}{\sqrt{1+x^3}-1}$$

Solution

We have

$$\sin(e^{x^3}-1)\sim e^{x^3}-1\sim x^3, \quad x\to 0^+$$

and

$$\sqrt{1+x^3}-1\sim \frac{1}{2}x^3, \quad x\to 0^+.$$

The given limit equals

$$\lim_{x\to 0} \frac{\sin(e^{x^3}-1)}{\sqrt{1+x^3}-1} = \lim_{x\to 0} \frac{x^3}{\frac{1}{2}x^3} = 2.$$

Exercise 1.39

Compute

$$\lim_{x\to 0^+} \frac{x^x - e^x + x^\alpha}{\sin\sqrt{x}} = L$$

as the parameter $\alpha > 0$ varies.

Solution

We have

$$\lim_{x\to 0^+} \frac{x\log x}{x^\alpha} = \begin{cases} 0, & \alpha < 1 \\ -\infty, & \alpha = 1 \end{cases}$$

hence, as $x \to 0^+$, the infinitesimal $x\log x$ is of lower order than x ($x = o(x\log x)$) but of higher order than x^α if $\alpha < 1$ ($x\log x = o(x^\alpha)$, $\alpha < 1$).
Hence, from the expansion

$$e^y = 1 + y + o(y),\ y \to 0,$$

it follows

$$x^x - e^x + x^\alpha = e^{x\log x} - e^x + x^\alpha =$$

$$1 + x\log x + o(x\log x) - 1 - x + o(x) + x^\alpha =$$

$$\begin{cases} x\log x + o(x\log x), & \alpha \geq 1 \\ x^\alpha + o(x^\alpha), & 0 < \alpha < 1. \end{cases}$$

The given limit equals

$$\lim_{x\to 0^+} \frac{x^x - e^x + x^\alpha}{\sin\sqrt{x}} = \begin{cases} \lim_{x\to 0^+} \frac{x\log x}{\sqrt{x}} = \lim_{x\to 0^+} \sqrt{x}\log x = 0, & \alpha \geq 1 \\ \lim_{x\to 0^+} \frac{x^\alpha}{\sqrt{x}} = 0, & 1/2 < \alpha < 1 \\ \lim_{x\to 0^+} \frac{\sqrt{x}}{\sqrt{x}} = 1, & \alpha = 1/2 \\ \lim_{x\to 0^+} \frac{x^\alpha}{\sqrt{x}} = +\infty, & 0 < \alpha < 1/2. \end{cases}$$